



Patient Profile

Saubhik Bhaumik, 27 years old, Male

Summary of Results

The complete blood chemistry panel is interpreted by the predictive model as **Normal**. All evaluated parameters fall within their established reference ranges, indicating no clinically significant hepatic or protein-metabolic abnormalities detected in this assessment.

Detailed Test Explanations

Serum Bilirubin (Total) measures overall bilirubin concentration, reflecting hepatic excretion and hemolysis. A value of 0.9 mg/dl is well within the normal 0–2.1 mg/dl range, suggesting no cholestasis or hemolytic burden.

Serum Bilirubin (Direct) identifies conjugated bilirubin; 0.2 mg/dl lies comfortably inside the 0–0.3 mg/dl reference, indicating normal hepatocellular conjugation.

Serum Bilirubin (Indirect) reflects unconjugated bilirubin; 0.7 mg/dl falls within 0–2.0 mg/dl, further supporting normal hemolytic and hepatic processing.

SGPT (ALT) is a liver transaminase sensitive to hepatocellular injury. A level of 36 U/L is within the normal 13–40 U/L, signifying no detectable hepatocellular damage.

SGOT (AST), another transaminase, is 32 U/L, also within the 0–37 U/L normal span, corroborating the ALT result.

Serum Alkaline Phosphatase data were not provided; therefore, hepatic cholestasis cannot be evaluated from this parameter.

Serum Protein, Total assesses overall plasma proteins. At 7.2 g/dl, the patient remains within the 6.4–8.3 g/dl range, indicating normal protein synthesis and nutritional status.

Serum Albumin reflects hepatic synthetic function and nutritional reserve; 4.7 g/dl is within 3.5–5.2 g/dl, consistent with adequate albumin production.

Globulin measures non-albumin proteins involved in immune and transport functions; 2.5 g/dl is within 1.8–3.6 g/dl, showing balanced globulin levels.

AIG Ratio (Albumin/Globulin ratio) is 1.88, comfortably within the 1.1–2.4 interval, supporting balanced protein synthesis and immune function.

Overall Interpretation

The laboratory profile demonstrates normal hepatic enzymology, bilirubin handling, and protein metabolism. The absence of any transaminase elevation or bilirubin derangement suggests no active liver injury or cholestatic process. The protein panel indicates adequate synthetic function and a healthy albumin/globulin balance. The predictive model's normal flag is consistent with these findings.

Recommendations

Maintain a balanced diet rich in fruits, vegetables, lean proteins, and adequate hydration to support liver health. Regular physical activity and moderation of alcohol intake further reduce future hepatic risk. Routine health screening every 1–2

years is advised, and any new symptoms such as jaundice, abdominal pain, or unexplained fatigue should prompt timely reevaluation.

Disclaimer

This report is informational and not a substitute for personalized medical advice. For clinical decisions, consult a qualified healthcare professional.